

LGE Scar Quantification Using Foundation Models for Cardiac Disease Classification



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Objective: Using cardiac MRI foundation model to improve scar segmentation and derive clinically useful metrics for disease classification

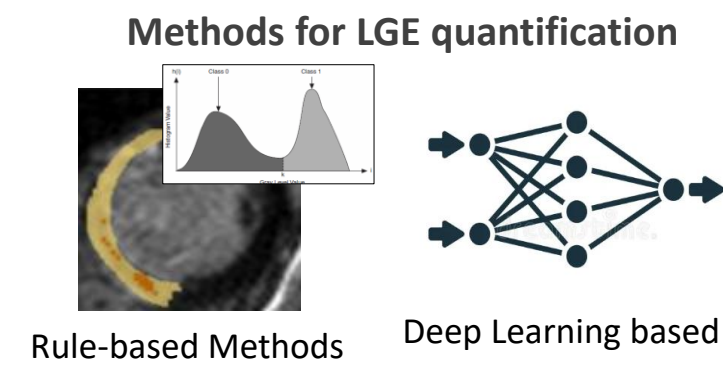
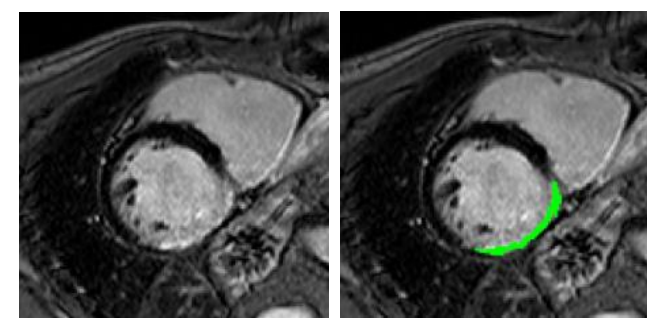
Introduction

Why LGE detection?

- Crucial for identifying myocardial fibrosis and scar
- Key diagnostic marker in a range of cardiomyopathies
- Strong prognostic implications in clinical decision-making

Challenges in developing LGE detection models

- Lack of reliable ground truth segmentation masks
- Manual annotation is time-consuming and expert-dependent



Data

Center 1: Algorithm Development Dataset: 212 patients

- Training: 159 patients (1516 images), Testing: 53 patients (497 images). 20% of the training set used for validation
- Segment-wise binary scar labels (based on the AHA 17-segment model) were annotated by an expert reader (1 = scar present, 0 = no scar)

Center 2: Disease Classification Dataset

- Independent cohort of 662 patients (different continent, healthcare system, sequence)
- DL model used as-is for scar segmentation
- Patients were labeled as : a) normal (NORM, 43%), b) dilated cardiomyopathy (DCM, 11%), c) hypertrophic cardiomyopathy (HCM, 13%); and d) ischemic heart disease (IHD, 33%)

Methods

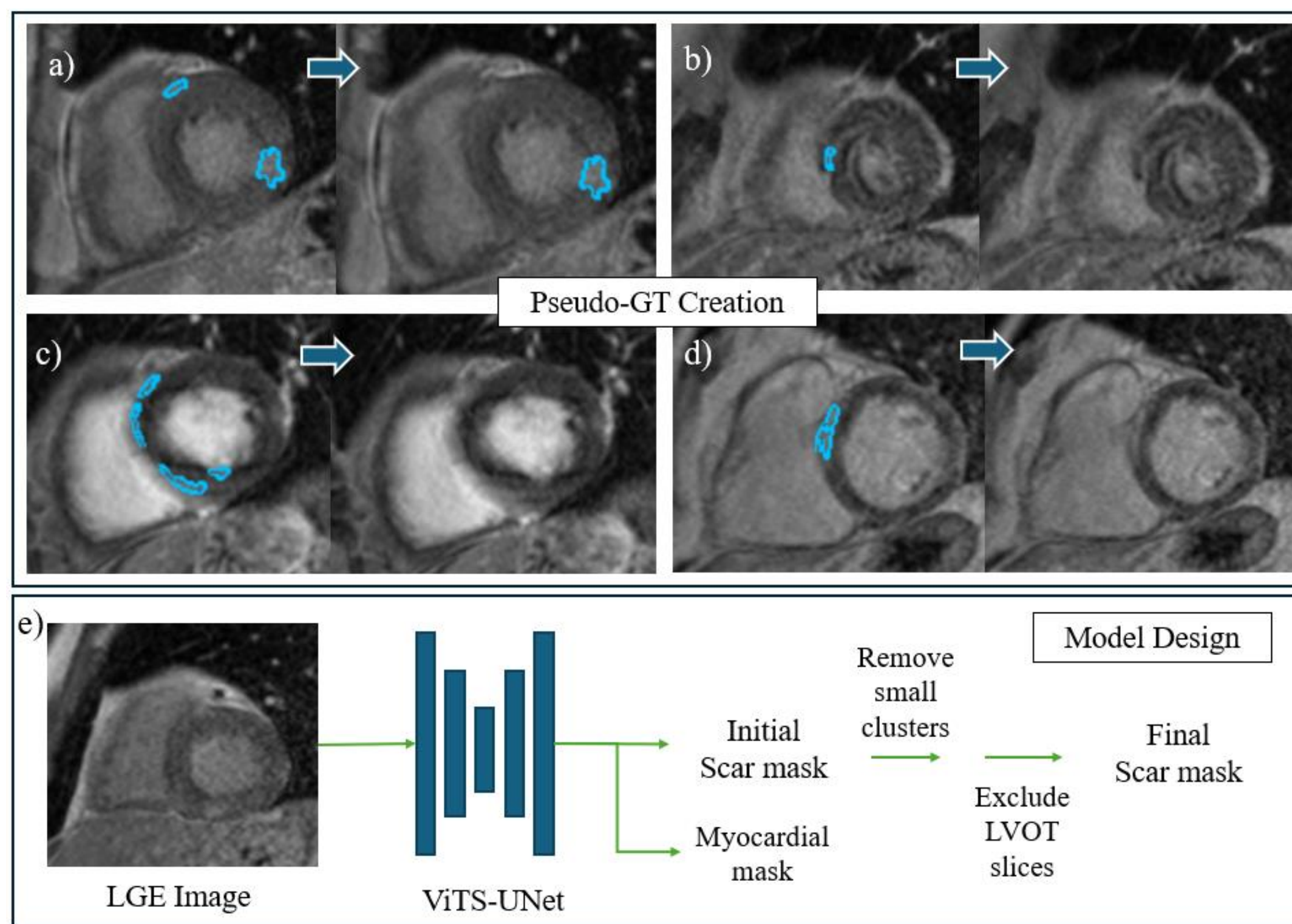


Fig. 1. (a-d) illustrate examples of initial masks from 1.5 std method that were refined to generate the pseudo-GT masks for training. (e) shows the pipeline used for inference.

Pseudo-GT generation on Algorithm Development Dataset:

- Initial masks: Created using n-std thresholding (n = 1.5) for maximum scar coverage.
- Problem: Overestimation; false positives (FP) in clinically normal segments.
- Refinement: Removed FP clusters in normal segments. Retained clusters in positive segments (true positives) to preserve sensitivity while improving specificity.

Proposed model:

- Architecture: U-Net transformer with ViT/8 encoder (CMR foundation model pretrained on 36M images using contrastive learning) [1].
- Input preprocessing: Resampled to 1x1 mm, center cropped to 224x224.
- Training: Jaccard loss, cosine annealing LR (0.001), geometric + elastic augmentations.
- Multi-task learning: Joint scar + myocardium segmentation (multi-channel outputs).
- Ensembling: Final predictions averaged from 3 models with different seeds.
- Post-processing: Removed small isolated clusters (<20 mm², >5 mm away from other scar); excluded slices above basal LV (identified with cine-based classifier + DICOM metadata mapping).

Validation

Patient-wise LGE Detection (Same-center)

- Metrics: Accuracy, balanced accuracy, AUC, sensitivity, specificity.
- LGE-positive: Patients with nonzero scar burden.

Disease Classification (External)

- Scar segmentation on LGE image stack, $Scar\ Burden = \frac{Scar\ area}{Myocardial\ area}$ per patient
- Scar burden added as an additional feature to 96 cine-based measurements[2].
- XG Boost Models for disease classification: Cine-only (96 features) vs. Cine+LGE (97 feats.),
- Evaluation: AUC, accuracy, feature importance (SHAP).

Results

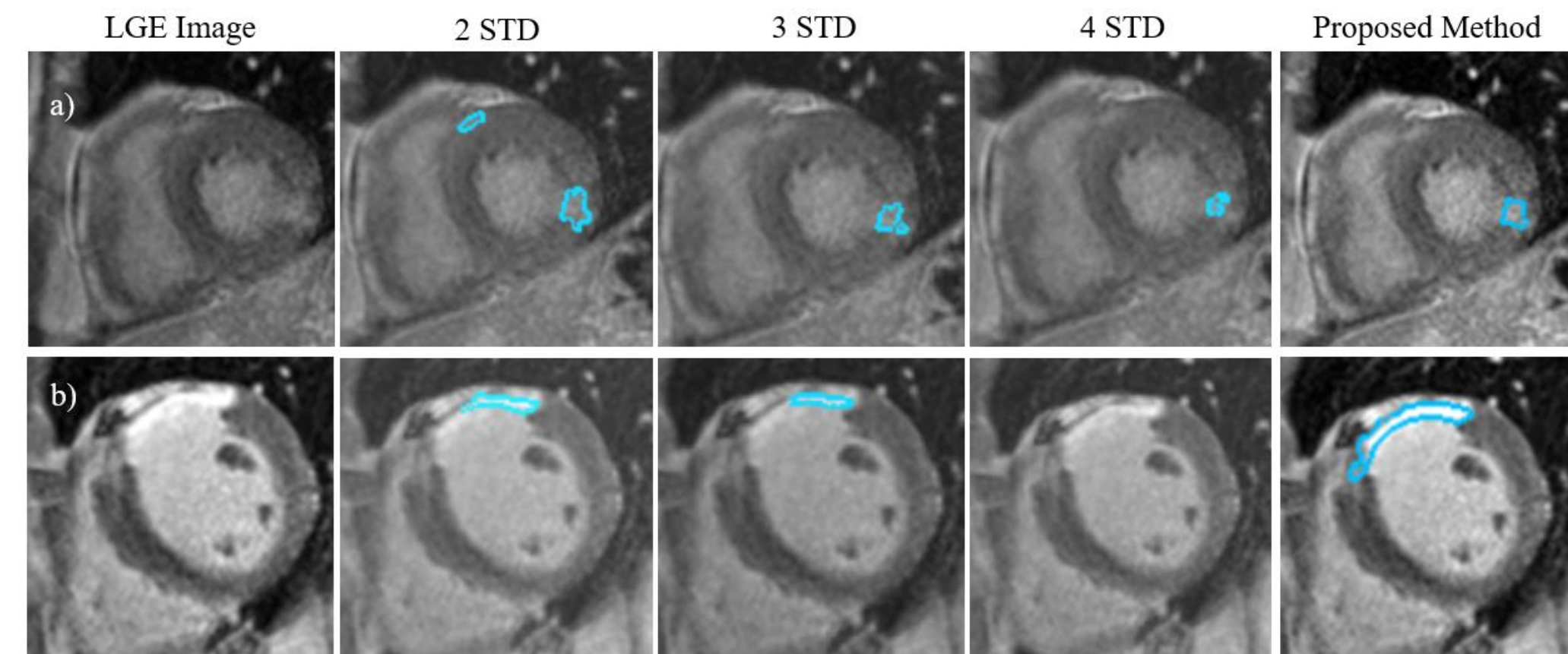


Fig. 2: Representative results on two patients from the same-center testing set. N-std deviation methods form the baseline comparisons.

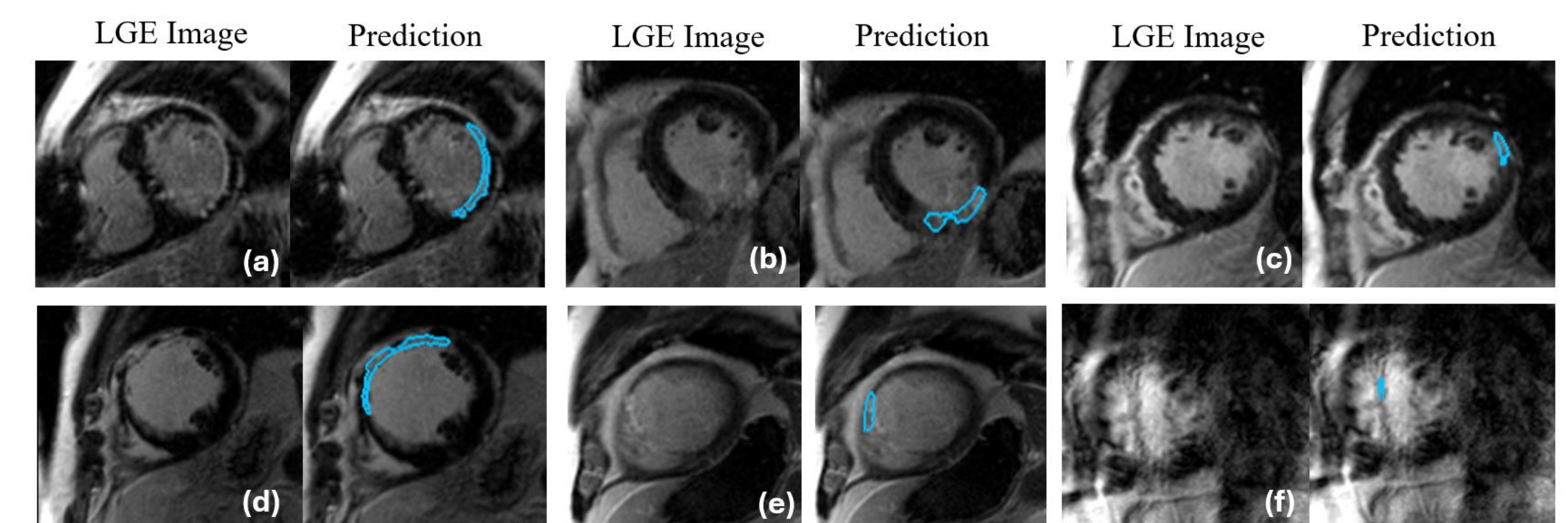


Fig. 3: Results on images from six patients on the external dataset. (a-d) True positive results, (e) Partial true positive, (f) False positive due to image artifact. Scar segmentation model is run as-is.

Table 1. Results on same-center testing set (53 patients). The highest and second-highest values are highlighted in bold and underlined, respectively. Confusion matrices are presented as [TN, FP], [FN, TP].

Method	ROC-AUC	Accuracy	Balanced Accuracy	Confusion Matrix	Sensitivity	Specificity
2-std	0.76	0.68	0.61	$\begin{bmatrix} 6 & 15 \\ 2 & 30 \end{bmatrix}$	0.94	0.29
3-std	0.76	0.70	0.68	$\begin{bmatrix} 13 & 8 \\ 8 & 24 \end{bmatrix}$	0.75	0.62
4-std	0.71	0.66	0.70	$\begin{bmatrix} 19 & 2 \\ 16 & 16 \end{bmatrix}$	0.50	0.91
Proposed	0.96	0.89	<u>0.86</u>	$\begin{bmatrix} 16 & 6 \\ 0 & 31 \end{bmatrix}$	1.00	0.73

Increasing n in the baseline thresholding algorithms results in a trade-off between sensitivity and specificity, with lower n values yielding higher sensitivity but lower specificity, and vice-versa. **The proposed model achieves the highest AUC and accuracy metrics.**

Table 2. Results on disease classification in external dataset (n = 131 patients). Incorporating scar burden as a feature enhances classification performance across all classes

Model	ROC-AUC				Weighted AUC	Accuracy
	NORM	DCM	HCM	IHD		
Cine-only	0.95	0.87	0.93	0.83	0.90	0.76
Cine+LGE	0.97	0.88	0.95	0.89	0.93	0.79

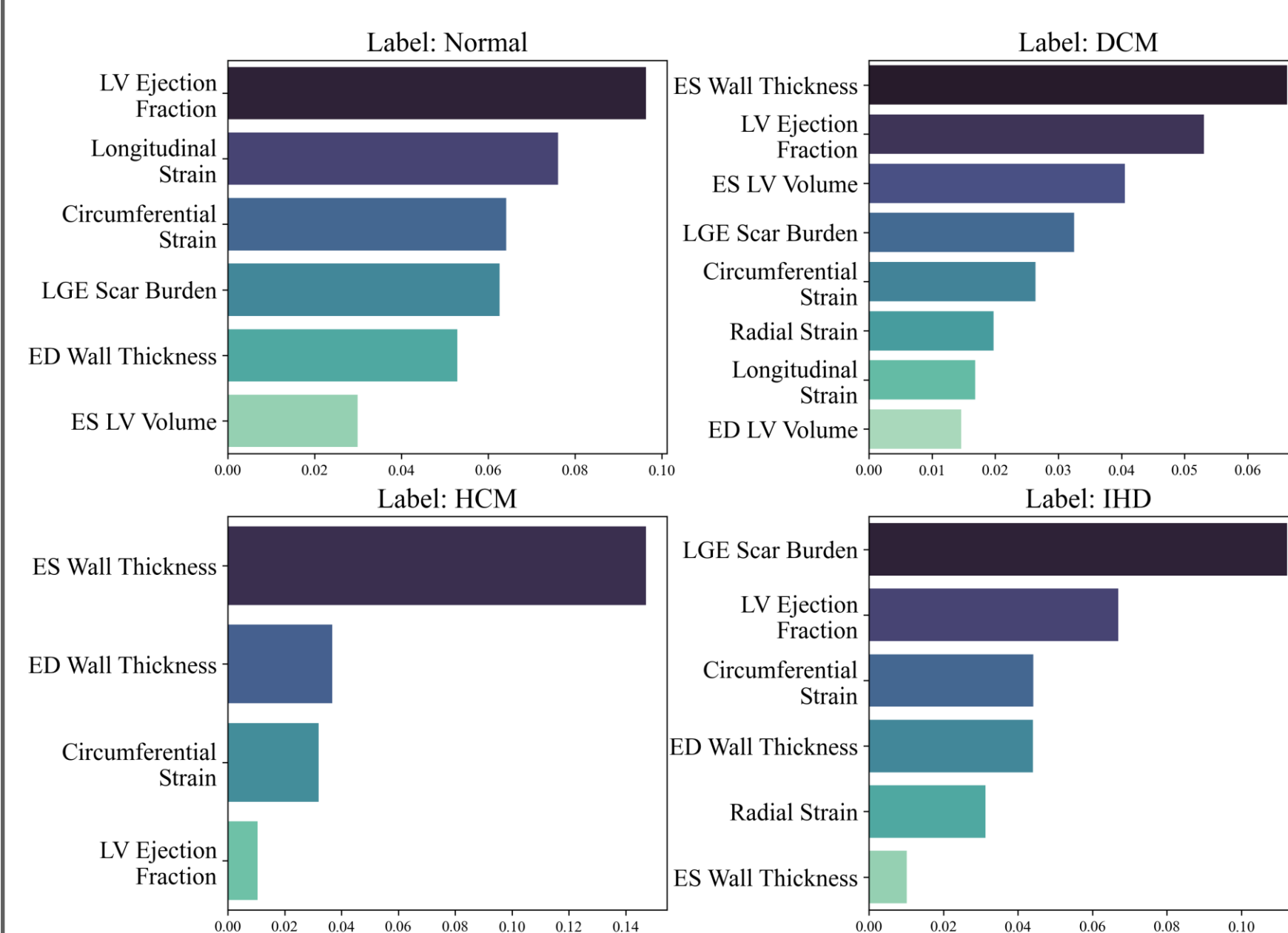


Fig. 4. Feature attributions for various disease classes using SHAP analyses. LGE scar burden is the most influential feature for distinguishing IHD while also contributing to NORM and DCM classification, consistent with clinical protocols.

Table 3. Ablation study: Effect of different encoders on performance. n = 53 patients. **Foundation model as encoder outperforms other methods.**

Encoder Type	ROC-AUC	Accuracy	Balanced Accuracy
CMR Foundation Model (ViT/8)	0.96	0.89	0.86
Convolutional encoder (Dense121) pretrained on related task	0.82	0.81	0.83
ViT/8 pretrained on ImageNet	0.89	0.75	0.71

Conclusions

- Developed deep learning method for LGE scar quantification with strong performance and generalizability, using a CMR specific foundation model as the encoder.
- CMR FM was trained on 36M multi-center, multi-sequence cardiac MR images
- Scar burden improved disease detection and interpretability, offering a scalable, annotation-efficient solution.
- Limitations: single-vendor data, no expert LGE masks, no ground truth in classification cohort.

[1] Jacob, A.J., Borgohain, I., Chitiboi, T., Sharma, P., Comaniciu, D., Rueckert, D.: Towards a vision foundation model for comprehensive assessment of cardiac mri. arXiv preprint arXiv:2410.01665 (2024)

[2] Jacob, A.J., Chitiboi, T., Schoepf, U.J., Sharma, P., Aldinger, J., Baker, C., Lautenschlager, C., Emrich, T., Varga-Szemes, A.: Deep-learning-based disease classification in patients undergoing cine cardiac mri. Journal of Magnetic Resonance Imaging (2024)

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